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KINETIC BEHAVIOR OF TRIGA REACTORS

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KINETIC BEHAVIOR OF TRIGA REACTORS

1. INTRODUCTION

The kinetic behavior of the TRIGA reactor has been discussed in previous reports. This report will review the recent experimental work related to the Advanced TRIGA Prototype Reactor (ATPR) program, with some emphasis on the ultimate potential of pulsing operation. The characteristics of the reactor which are responsible for its unique pulsing capabilities will be examined and the methods for calculating the parameters necessary to predict the pulsing behavior of TRIGA reactors will be discussed. In conclusion, some comparisons of experimental and calculated results will be made.

2. ATPR EXPERIMENTAL AND OPERATIONAL HISTORY

Research and development activities with the ATPR have reached the point where \$5 step reactivity insertions have been tested. A \$5 pulse in a core of 100 fuel elements released an energy of 54 Mw-sec., achieved a peak power of 8400 Mw, a pulse width of 5.5 msec. and a peak fuel temperature of 1050°C. Since the energy release in a pulse is a function of the prompt reactivity insertion, the \$5 experiments in the ATPR have increased the tested performance of this system to a factor of approximately two over the routine \$3 pulses characteristic of the many TRIGAs in operation throughout the world. Performance data for \$5 and \$3 pulses are compared with calculated values in a table at the end of this paper.

Experience with the ATPR core has now included over 7000 pulses and some 4000 Mw-hr of steady state operation at powers up to 1.5 Mw (th). Over 1000 of the pulses involved reactivity insertions of \$4 or more. This continuing testing program is designed to increase the understanding of the kinetic behavior of the system in the upper temperature ranges by providing metallurgical and engineering data as well as simply extending the reactor performance.

3. TRIGA FUEL DESCRIPTION

Fuel elements in the ATPR core are stainless steel clad U-ZrH_{1.7} and the core water volume fraction is 1/3. The uranium loading (enriched to 20% U²³⁵) in the fuel is between 8 and 8.5 wt-%. Having a H/Zr ratio > 1.5 is advantageous because these higher-hydride compositions are single phase (delta) well above 1000°C and are not subject to phase separation on thermal cycling. Lower hydrides, which are used in many lower power TRIGA cores, have a substantial volume change

associated with phase transformations at approximately 530°C . This phase and volume change coupled with temperature gradients through the fuel element can cause distortion or bending in fuel elements if the lower hydride is used at temperatures above $\sim 530^{\circ}\text{C}$. A phase diagram for zirconium hydride is shown in Figure 1. Hydrides of 1.65 to 1.7, substantially removed from the phase boundary at ~ 1.5 (61.4 atom percent), were chosen for the original high performance TRIGA elements to assure the single phase composition even under extreme temperature gradient and heating rate conditions.

A potential limitation on the maximum fuel temperature is the possible diffusion of hydrogen from the fuel with a resulting pressure within microscopic voids in the fuel body and in gas space in the fuel can. Figure 2 gives values for the dissociation pressure isochores of zirconium hydride, showing the exponential increases in the pressure with advancing fuel temperature. Also to be noted is the significant pressure advantage to be gained by using low H/Zr ratios. Extensive experience with the U-ZrH_{1.7} fuel in the ATPR, along with many tests on other H/Zr ratios indicates that H/Zr ratios approaching 1.55 could allow pulsing operation in a regime that is above the phase boundary (H/Zr = ~ 1.5), even under conditions of hydrogen migration, and yet would allow a maximization of the temperatures to which the fuel could be pulsed under any possible pressure limits.

The pressure-temperature relationships shown in Figure 2 represent equilibrium hydrogen pressure for steady state conditions. Greatly reduced pressures are recorded outside the fuel matrix for a given temperature when the temperature is reached for only a short period of time (< 1 second) during a pulse. In some very recent experiments, measurements have been made of the internal pressure in the fuel can for pulses yielding a local peak temperature of $\sim 1100^{\circ}\text{C}$ (calculated). The resulting pressure rise was only ~ 24 psi compared with 575 psi predicted from the equilibrium steady state data. Out of core (laboratory) experiments on U-ZrH_{1.7} fuel sections have confirmed the lower hydrogen pressures for transient temperatures. They further show that the transient hydrogen pressure measured in the fuel can is a function of heating rate and that with high heating rates, pressures much reduced from the equilibrium values are obtained. These experiments involved the use of a few fuel elements that were more highly loaded in U²³⁵ than the rest of the elements in the core, and for this reason these fuel elements significantly led the rest of the core in temperature reached for a given pulse size.

4. HEAT TRANSFER CONSIDERATIONS

Detailed calculations have been made in order to understand more fully the temperature distribution within a TRIGA fuel element during and after a pulse. The temperature-time distribution throughout a fuel body

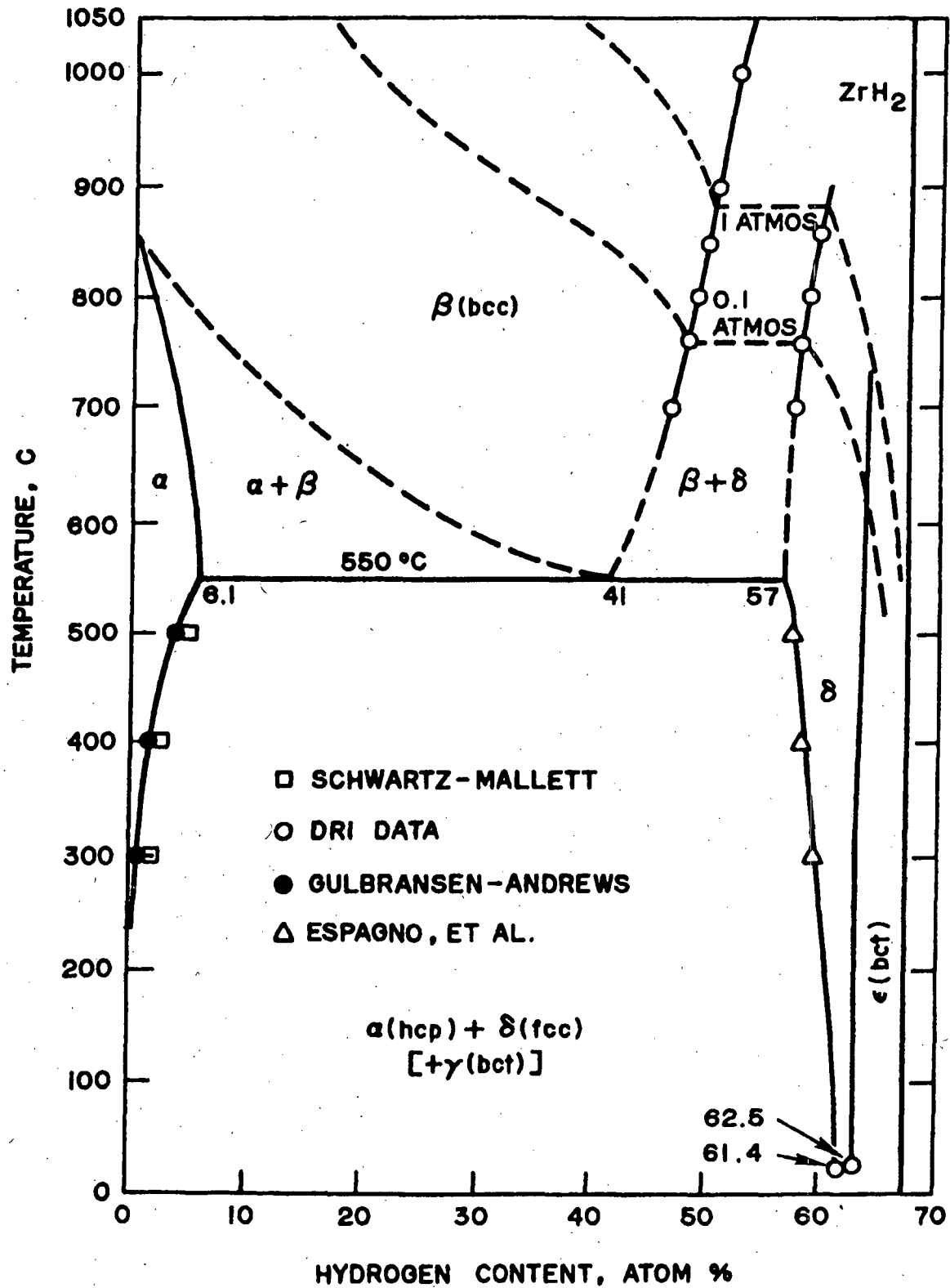


Fig. 1—Zirconium hydrogen phase diagram⁽¹⁾

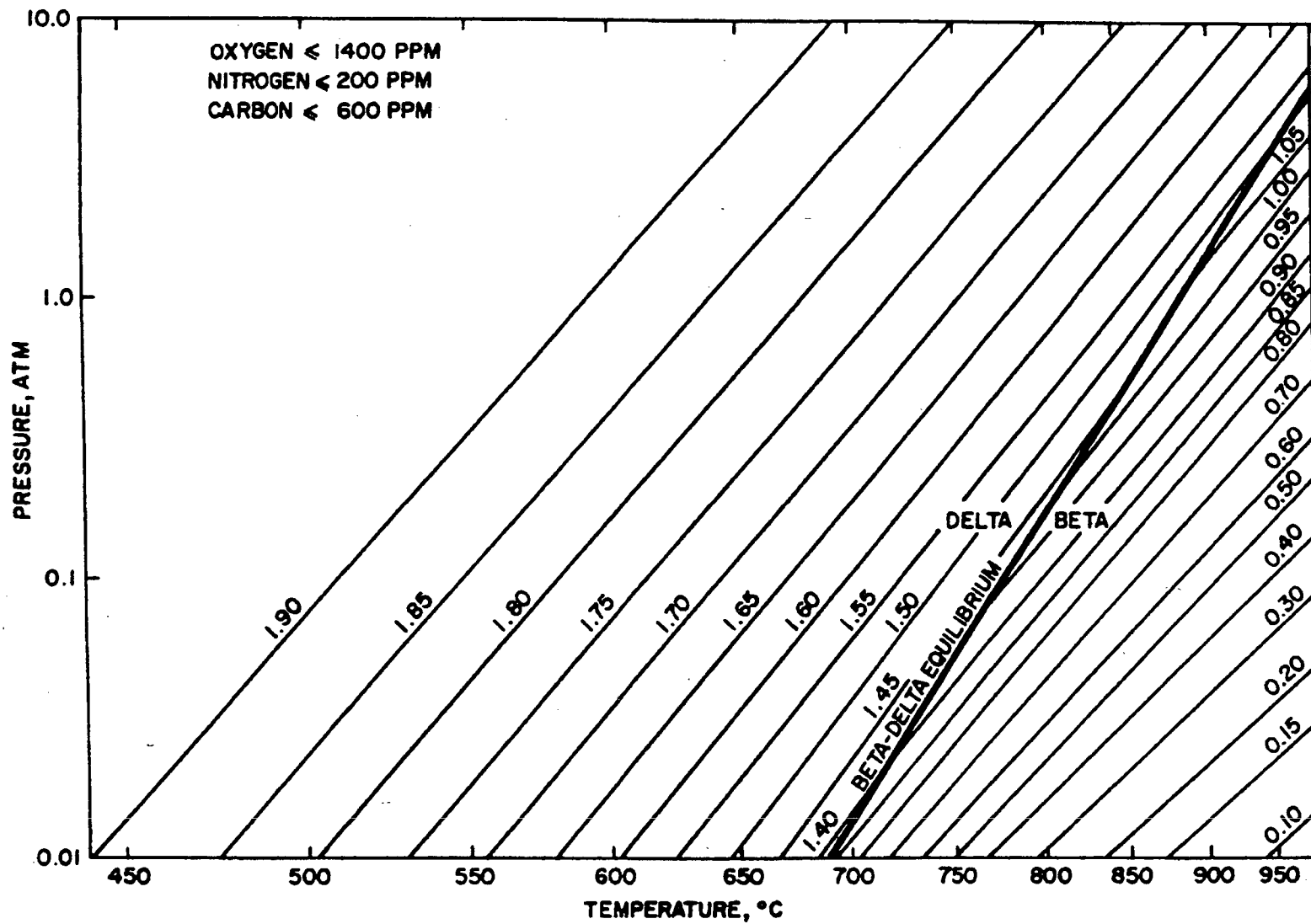


Fig. 2—Dissociation pressure isochores of zirconium hydride
(expressed as H/Zr atom ratios)⁽²⁾

was derived from computer calculations for the case of a maximum fuel temperature of 750°C and is shown in Figure 3. A somewhat artificial boundary condition was imposed for these calculations, namely, that only nucleate boiling will occur at the water-clad interface regardless of the heat flow. It was further assumed that there was no thermal resistance at the fuel-clad interface. These assumptions are conservative in the sense that they lead to calculations of a maximum value of heat flux and a maximum cooling rate. This does not change the peak fuel temperature. Based on the above calculations, Figure 4 shows a composite of the time change in the heat flux, clad temperature, and water coolant temperature following a pulse.

Experimental evidence and thermal analysis indicate that film boiling occurs around the hottest elements for pulses approaching \$5. This is a condition where the element is temporarily surrounded by an insulating sheath of steam. This boiling is caused by rapid heat transfer to the coolant which results from the large initial temperature gradient between the clad and the fuel body. The nuclear pulse heating of the fuel is essentially completed before heat transfer begins to occur. Subsequently, the fuel body can only cool down, and consequently the fuel temperature constitutes an upper bound to the clad temperature. If the cladding temperature reached 800°C the rupture pressure for the cladding would be 450 psi, which is well above even the equilibrium hydrogen gas pressure. Since this is safe, the significance of film boiling is limited to its effects upon useful clad life. (This is in sharp contrast to the hazardous implications associated with film boiling in steady state reactors in which fuel temperatures can continue to rise.) Since only after-heat is being generated in the fuel after completion of the pulse, the driving force for the heat transfer rate soon disappears, as does the film boiling. In order to further increase the design temperature for the pulse fuel without decreasing clad life, a modified element was designed and tailored for high-temperature pulse performance. These elements incorporate a built-in gap between the fuel and clad which adequately reduces the peak heat transfer rates to the coolant, so that film boiling is precluded. The validity of this design approach was substantiated by results of the tests mentioned previously; in these tests 5 elements of gapped design were tested extensively.

The new gapped element design will be used in an annular core reactor now under construction with a 9-inch diameter dry central irradiation region, where integrated fast neutron fluxes $>10^{15}$ nvt per pulse will be routinely achieved. This system is designed to operate at peak fuel temperatures of 1000°C , with a core energy release of 106 Mw-sec, and a peak power of 20, 300 Mw.

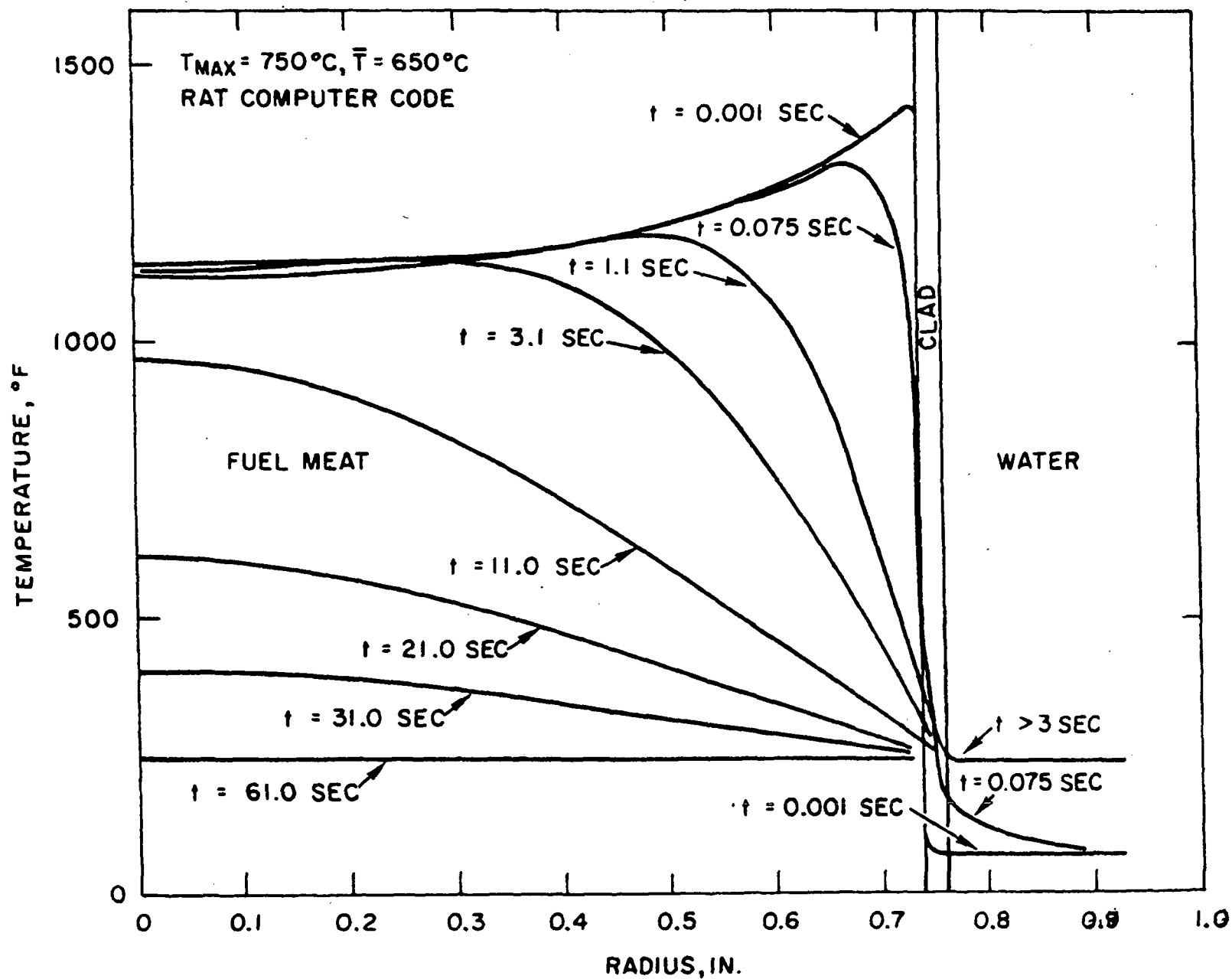


Fig. 3—Temperature distribution in fuel-moderator material following pulse

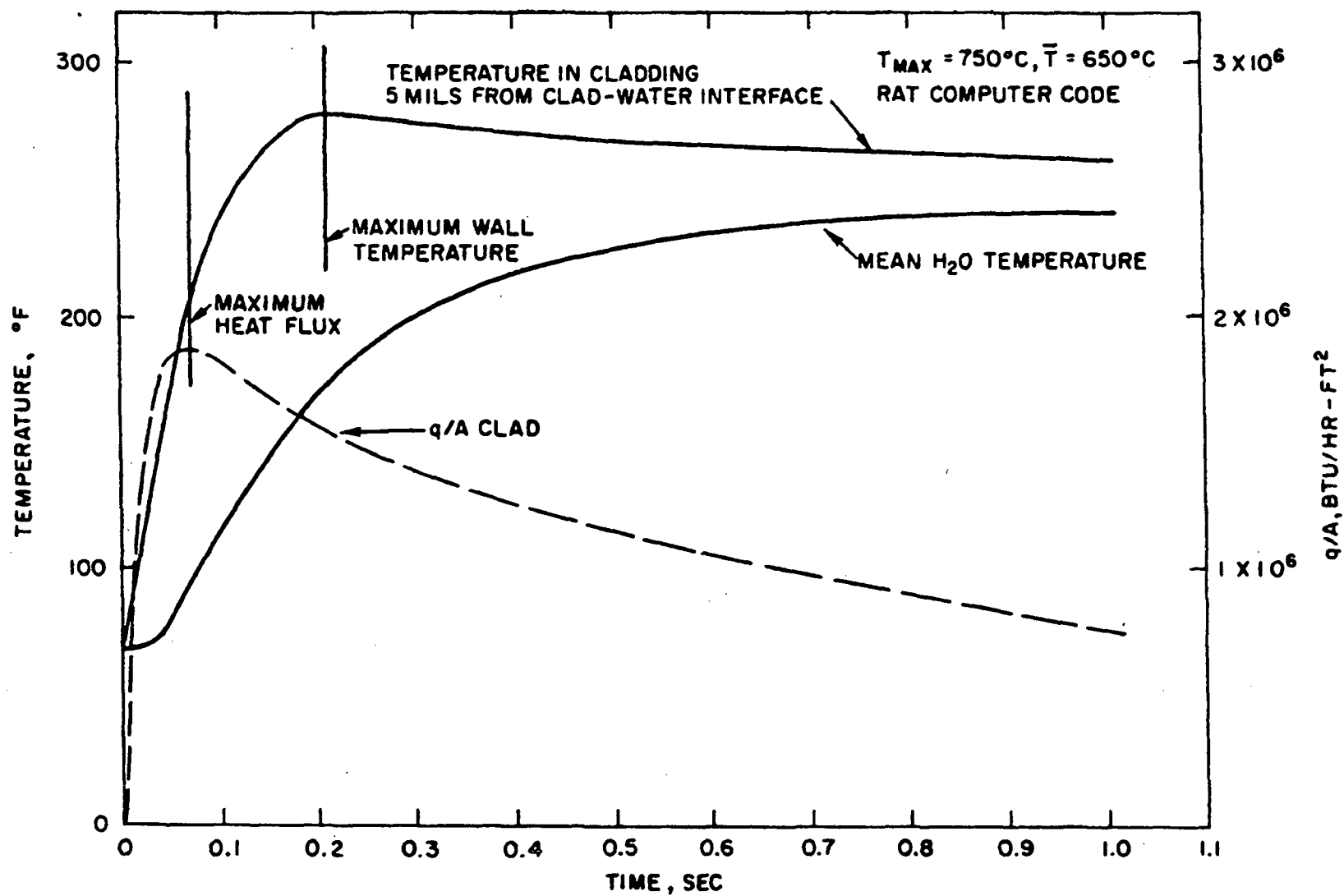


Fig. 4—Heat flux, clad temperature, and water temperature (zero flow) following pulse

5. TEMPERATURE COEFFICIENT

The prompt negative temperature coefficient is the basic characteristic of a TRIGA core which determines its behavior during pulse operation. This temperature coefficient arises primarily from a change in the disadvantage factor resulting from the heating of the uranium-zirconium-hydride fuel-moderator elements. The coefficient is prompt because the fuel is intimately mixed with a large portion of the moderator and thus fuel and solid moderator temperatures rise simultaneously. A quantitative calculation of the temperature coefficient requires a knowledge of the energy dependent distribution of thermal neutron flux in the reactor.

The basic physical processes which occur when the fuel-moderator elements are heated can be described as follows: the rise in temperature of the hydride increases the probability that a thermal neutron in the fuel element will gain energy from an excited state of an oscillating hydrogen atom in the lattice. As the neutrons gain energy from the ZrH their mean free path is increased appreciably. This is shown qualitatively in Figure 5. Since the average chord length in the fuel element is comparable with a mean free path, the probability of escape from the fuel element before capture is increased. In the water the neutrons are rapidly re-thermalized so that the capture and escape probabilities are relatively insensitive to the energy with which the neutron enters the water. The heating of the moderator mixed with the fuel thus causes the spectrum to harden more in the fuel than in the water. As a result, there is a temperature dependent disadvantage factor for the unit cell in the core which decreases the ratio of absorptions in the fuel to total-cell absorptions as the fuel element temperature is increased. This brings about a shift in the core neutron balance giving a loss of reactivity.

The temperature coefficient then, depends on spatial variations of the thermal neutron spectrum over distances of the order of a mean free path with large changes of mean free path occurring because of the energy change in a single collision. A quantitative description of these processes requires a knowledge of the differential slow neutron energy transfer cross section in water and zirconium hydride, the energy dependence of the transport cross section of hydrogen as bound in water and zirconium hydride, the energy dependence of the capture and fission cross sections of all relevant materials, and a multigroup transport theory reactor description which allows for the coupling of groups by speeding up as well as by slowing down.

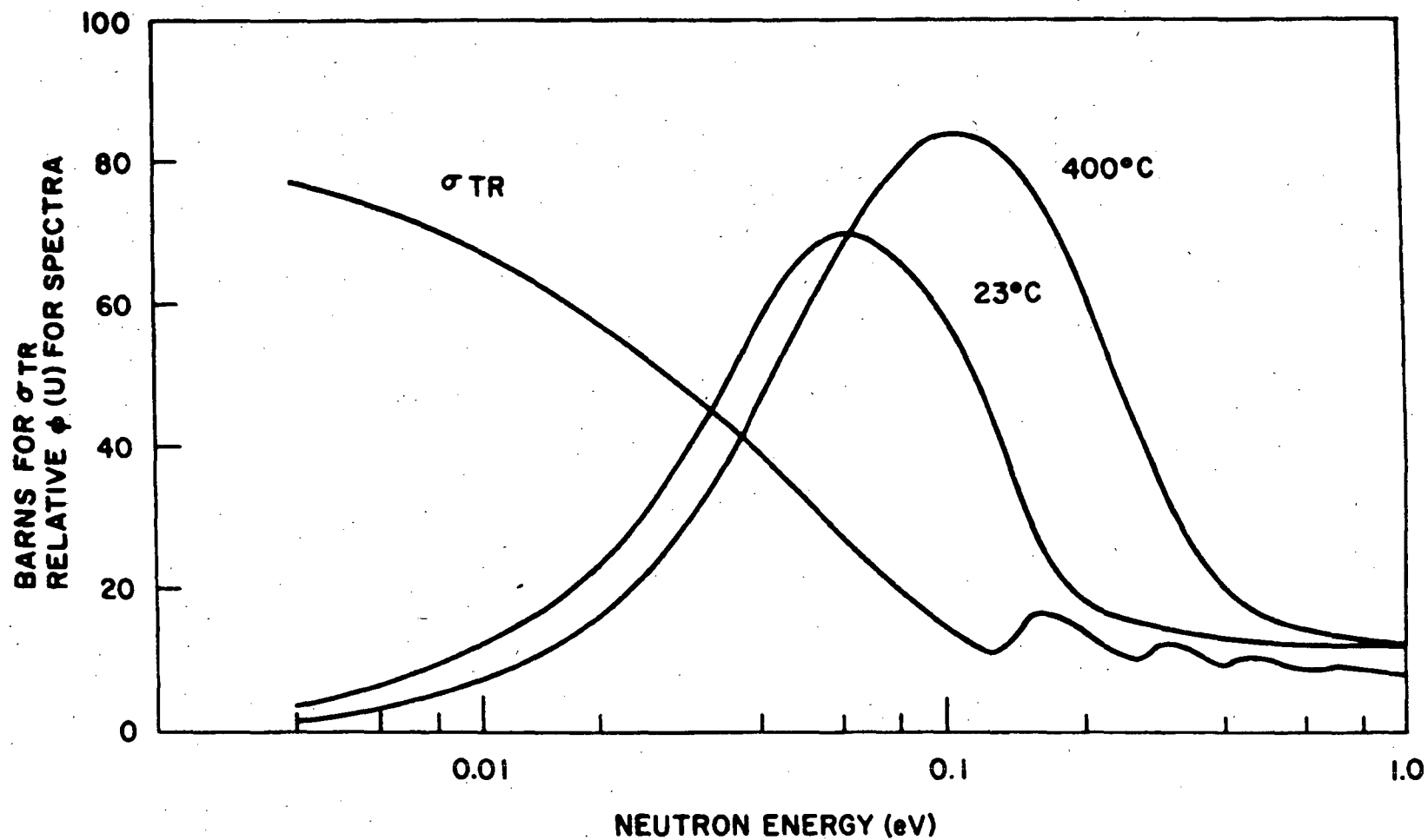


Fig. 5—Transport cross section for hydrogen in zirconium hydride and average spectra in TRIGA $ZrH_{1.7}$ fuel element for 23°C and 400°C fuel

5.1. Codes Used for Calculations

Recent calculational work on the temperature coefficient has utilized a group of General Atomic-developed codes, GAM-II,⁽³⁾ GATHER-II,⁽⁴⁾ GAZE-2,⁽⁵⁾ and GAMBLE,⁽⁶⁾ as well as DTF-IV,⁽⁷⁾ an S_n multigroup transport code written at Los Alamos. Neutron cross sections for energies above thermal (>1 eV) were generated by the GAM-II code where fine group (~ 100) cross sections, stored on tape for all commonly used isotopes, are averaged over a space independent flux derived by solution of the B_1 equations for each discrete reactor region composition. This code and its related cross section library predict the age of each of the common moderating materials to within a few percent of the experimentally determined values and utilizes the resonance integral work of Nordheim⁽⁸⁾ to generate cross sections for resonance materials which are properly averaged over the region spectrum.

Thermal cross sections were obtained in essentially the same manner using the GATHER-II code. However, scattering kernels were used to describe properly the interactions of the neutrons with the chemically bound moderator atoms. The bound hydrogen kernels used for hydrogen in the water were generated by the THERMIDOR code,⁽⁹⁾ using the thermalization work of Nelkin.⁽¹⁰⁾ Early thermalization work by McReynolds, Nelkin, Rosenbluth, and Whittemore⁽¹¹⁾ on zirconium hydride has been greatly extended at General Atomic⁽¹²⁾ and recent work by Parks has resulted in the SUMMIT⁽¹³⁾ code, which was used to generate the kernels for hydrogen as bound in ZrH. These scattering models have been used to adequately predict the water and hydride (temperature dependent) spectra as measured at the General Atomic linear accelerator.^(14, 15) Figure 6 illustrates the agreement between calculations and experiments for thermal neutron spectra in ZrH.

5.1.1. ZrH Model

Qualitatively, the scattering of slow neutrons by zirconium hydride can be described by a model in which the hydrogen atom motion is treated as an isotropic harmonic oscillator with energy transfer quantized in multiples of ~ 0.14 eV. More precisely, the SUMMIT model uses a frequency spectrum with two branches—one for the optical modes for energy transfer with the bound proton, and the other for the acoustical modes for energy transfer with the lattice as a whole. The optical modes are represented as a broad frequency band centered at 0.14 eV, and whose width is adjusted to fit the cross section data of Woods.⁽¹⁶⁾ The low frequency acoustical modes are assumed to have a Debye spectrum with a cutoff of 0.02 eV and a weight determined by an effective mass of 360.

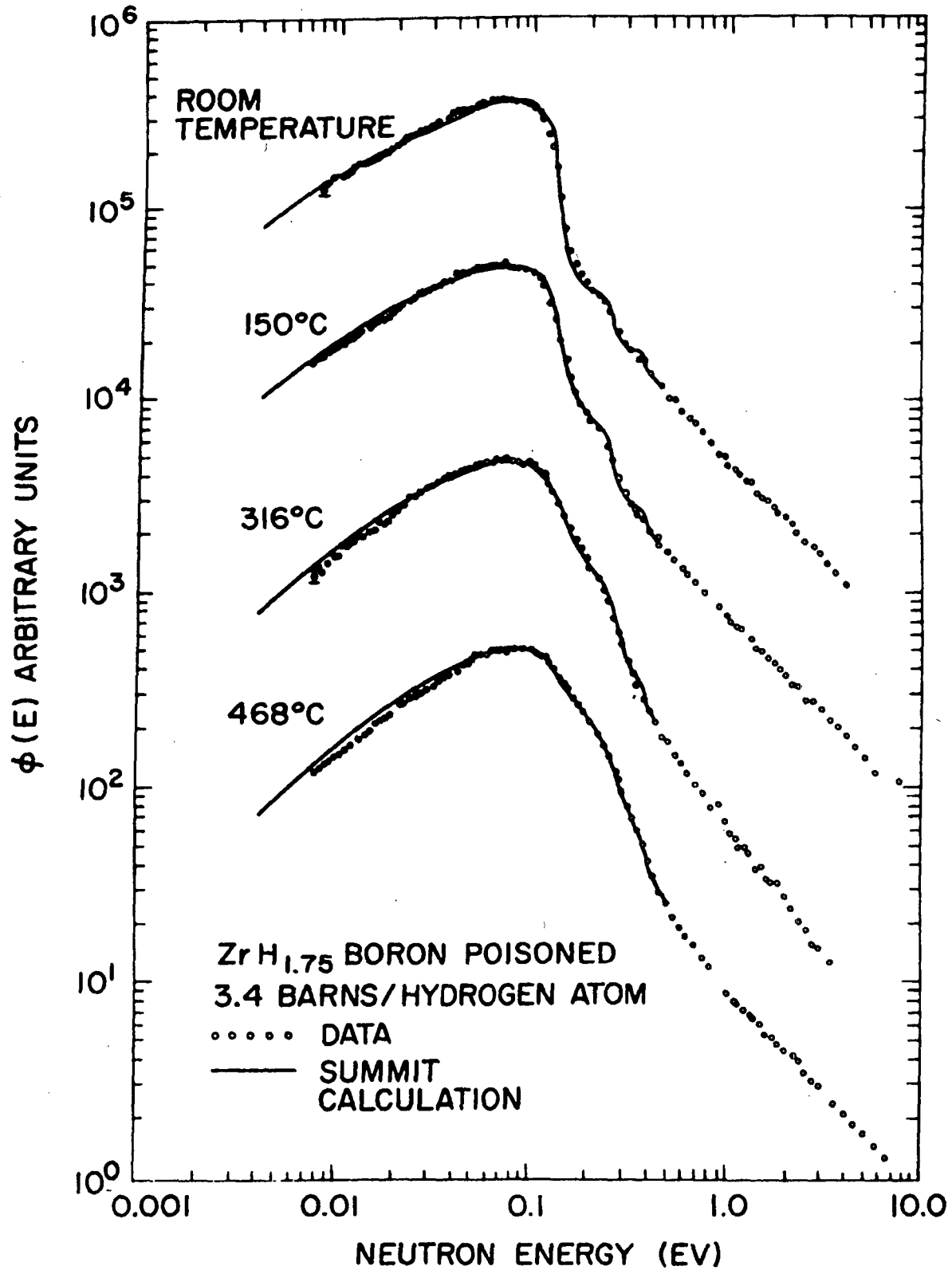


Fig. 6—Experimental and theoretical neutron spectra from ZrH_{1.75} showing the effect of temperature variation calculations done with SUMMIT code

This structure then allows a neutron to slow down by the transition in energy units of ~ 0.14 eV as long as its energy is above 0.14 eV. Below 0.14 eV the neutron can still lose energy by the inefficient process of exciting acoustic Debye-type modes in which the hydrogen atoms move in phase with the zirconium atoms which in turn move in phase with one another. These modes therefore correspond to the motion of a group of atoms whose mass is much greater than that of hydrogen, and indeed even greater than the mass of zirconium. Because of the large effective mass these modes are very inefficient for thermalizing neutrons, but for neutron energies below 0.14 eV they provide the only mechanism for neutron slowing down. (In a TRIGA core, the water provides for ample neutron thermalization below 0.14 eV). In addition, in the ZrH it is possible for a neutron to gain one or more energy units of ~ 0.14 eV in one or several scatterings, from excited Einstein oscillators. Since the number of excited oscillators present in a ZrH lattice increases with temperature, this process of neutron speeding up is strongly temperature dependent and plays an important role in the behavior of ZrH moderated reactors.

5.2. Calculations

The prompt negative temperature coefficient of TRIGA-type reactors is divided into four components for analysis. These components, with a description of the process producing the effect are as follows:

1. Cell—increased disadvantage factor with increased fuel temperature leading to a degradation in neutron economy.
2. Irregularities in the fuel lattice due to control rod positions—essentially same effect as (1) above; temperature dependent disadvantage factors occur in the cells where control rods are placed. These cells can contain water, water with an inserted control rod, or water with a control rod follower of aluminum (or aluminum with a central boron spine) to eliminate flux peaking adjacent to a withdrawn control rod.
3. Doppler broadening of U^{238} resonances—increased resonance capture with increased fuel temperature.
4. Leakage—increased loss of thermal neutrons from the core when the fuel is heated.

Calculations of the temperature coefficient are done in the following steps:

1. Multigroup cross sections are generated by the GAM-II and GATHER-II codes for a homogenized unit cell (1/3 water,

2/3 clad fuel-moderator element). Separate cross section sets are generated for each fuel element temperature by use of the temperature dependent hydride kernels (room temperature water is used for all prompt coefficient calculations) and Doppler broadening of the U^{238} resonance integral to reflect the proper temperature.

2. A k_{∞} is computed for each fuel element temperature by transport cell calculations, using the P_1 approximation. Comparisons have shown S_4 and S_8 results to be nearly identical. Group dependent disadvantage factors are calculated for each cell region (fuel, clad, and water) where the disadvantage factor is defined as $\bar{\phi}_g(\text{region})/\bar{\phi}_g(\text{cell})$.
3. The thermal group disadvantage factors are used as input for a second GATHER-II calculation where cross sections for a homogenized core are generated which give the same neutron balance (η_{thermal}) as the thermal group portion of the discrete cell calculation. The fast group disadvantage factors are small and have not been used; no second GAM-II calculation is done.
4. The core inhomogenieties are treated at this point. A large cell calculation is performed (1/4 core size if the core contains 4 rods) with the given control cell geometry discretely described and surrounded by a portion of the core represented by the cross sections from Step 3.
5. Disadvantage factors from (4) are compared with those from (3) and a third GATHER-II calculation is performed to produce cross sections which give a thermal neutron balance characteristic of the core structure and temperature. These cross sections are combined with the original fast group cross sections to give a full energy range set.
6. The cross sections from (5), for an equivalent homogenized core, are used in a full reactor calculation to determine the contribution to the temperature coefficient due to the increased leakage of thermal neutrons into the reflector with increasing hydride temperature. This calculation still requires several thermal groups, but transport effects are no longer of major concern. Thus, reactivity calculations as a function of fuel element temperature have been done on the entire reactor with the use of diffusion theory codes.

5.2.1. Cell Effect

The cell effect is determined from the thermal neutron balance calculated by the transport cell problem. It is due entirely to the change in thermal utilization (η) caused by heating of the hydride, but for ease of calculation it is computed as

$$\frac{1}{\eta} \frac{\Delta \eta}{\Delta T}.$$

Table I shows how the absorptions change with fuel element temperature for the various cell materials. Typical changes in the disadvantage factors which cause the shift in neutron balance are given in Table II. Power distributions within the cell are shown as a function of temperature in Figure 7 for a high hydride TRIGA fuel element.

Characteristic values for the cell component of the prompt negative temperature coefficient are $5.7 \times 10^{-5}/^{\circ}\text{C}$ for an 8.5 wt-% U (20% enriched)- $\text{ZrH}_{1.65}$ element and $3.5 \times 10^{-5}/^{\circ}\text{C}$ for an 8.0 wt-% U (20% enriched)- $\text{ZrH}_{1.0}$ element, averaged over a temperature range from 23° to 400°C . The increased cell effect for the higher hydride core is due in largest part to the substitution of stainless steel cladding for aluminum rather than the change in hydride. The stainless steel, with its greater absorption cross section, acts as a more effective barrier for neutrons which are speeded up and escape from the fuel-moderating material and are rapidly re-thermalized in the core water. The presence of an increased hydrogen density in the $\text{ZrH}_{1.65}$ also adds to the cell effect, but this contribution is only $\sim 20\%$ of the calculated increase. It is of interest to note here that the infinite medium negative temperature coefficient for a homogenized composition equal to that of a unit cell, but with no cell structure, is $2.1 \times 10^{-5}/^{\circ}\text{C}$ for high hydride over the 23° to 400°C temperature range. Thus, the cell effect is about three times as great as the temperature coefficient would be if calculated for a homogenized mixture of unstructured cell where the coefficient is due strictly to the effect of spectrum shift on the non- $1/v$ nature of absorptions in U^{235} .

Extensive survey calculations have shown the cell effect to be a strong function of the U^{235} concentration in the fuel and the volume fraction of water in the cell. As examples; if the uranium wt-% is increased from 8.5 to 12 in a $\text{ZrH}_{1.65}$ element, the coefficient is ~ 0.75 times the original value and increasing the enrichment of the 8 wt-% uranium from 20% to 93% cuts the coefficient by about a factor of 3. This is caused by the shortening of the mean free path in the fuel element as the U^{235} concentration is increased, making it more difficult for speeded up neutrons to escape from the fuel. If more water is added to the cell, allowing larger changes in the disadvantage factor for a given temperature change, the temperature coefficient can be increased by about 12% for a water fraction increase to 0.33 from 0.27. Increases in temperature coefficient are larger as the water volume fraction increases.

Table I

NEUTRON BALANCE VERSUS FUEL-MODERATOR TEMPERATURE
FOR FOUR THERMAL GROUPS OF TRIGA CELL CALCULATION

(Cell description: 2/3 stainless steel-clad 8.5 wt-%
U(20% enriched)-ZrH_{1.65}; 1/3 water)

Material	Total Absorptions	
	Cold Cell	Hot Cell
	Fuel-moderator (23°C) Water (23°C)	Fuel-moderator (400°C) Water (23°C)
H (ZrH)	0.0839	0.0840
Zr	0.0321	0.0322
U ²³⁵	0.7250	0.7134
U ²³⁸	0.0121	0.0121
SS	0.0801	0.0853
H (H ₂ O)	0.0668	0.0730
Oxygen	<u>0.0000</u>	<u>0.0000</u>
TOTAL	1.0000	1.0000

Table II
DISADVANTAGE FACTORS FOR A TRIGA CELL VERSUS
FUEL-MODERATOR TEMPERATURE

Energy Range (eV)	Fuel (23°C)	Clad	Water (23°C)	Fuel (400°C)	Clad	Water (23°C)
1.125 to 0.42	0.990	1.007	1.018	1.008	0.993	0.986
0.42 to 0.14	0.985	1.008	1.027	1.112	0.925	0.796
0.14 to 0.05	0.992	0.995	1.016	0.949	1.035	1.093
0.05 to 0.002	0.850	1.122	1.270	0.746	1.195	1.459

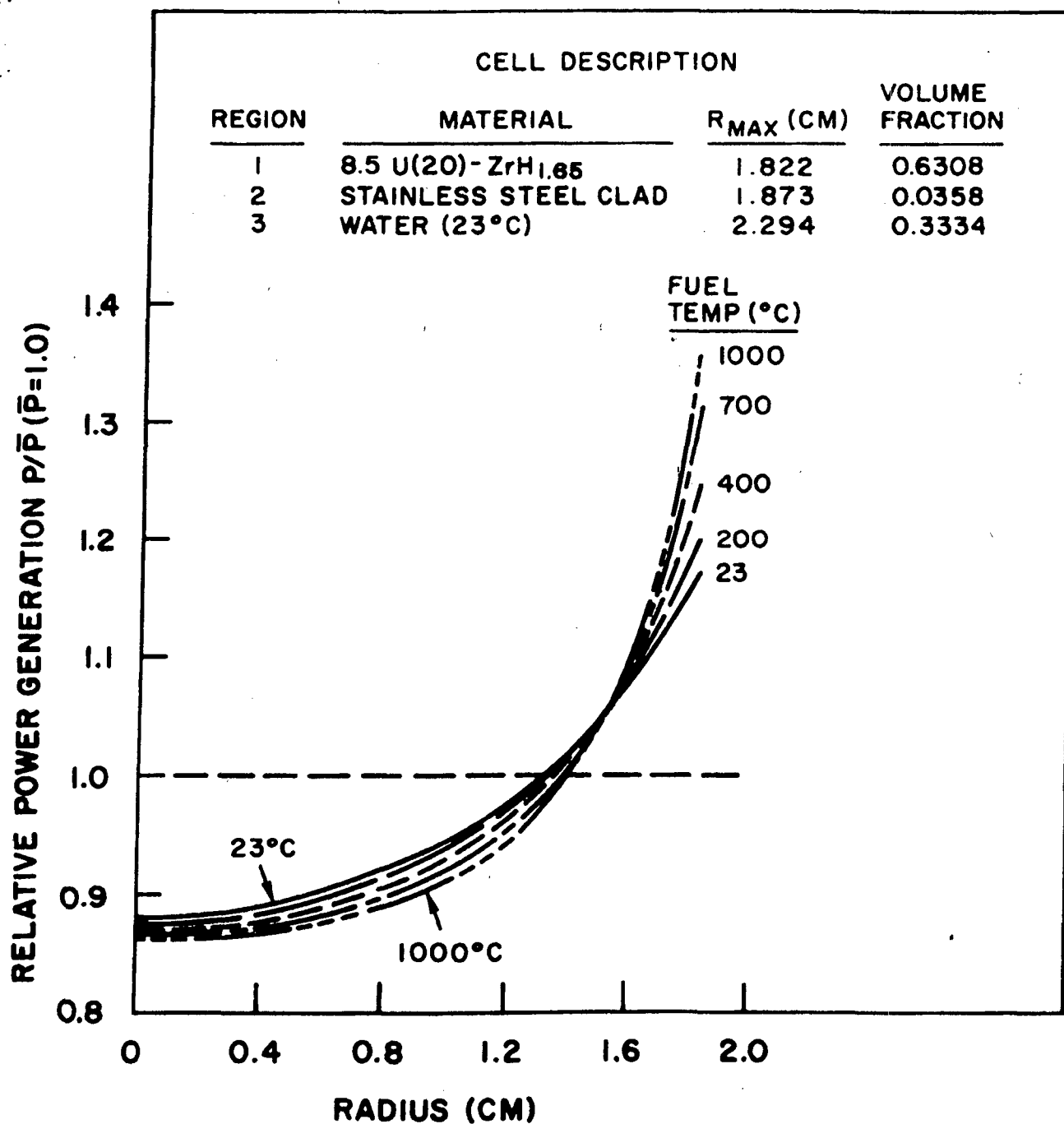


Fig. 7—Relative power generation in TRIGA cell versus fuel temperature

5.2.2. Group Structure

Values computed for the cell, and leakage effects, are somewhat dependent on the number of groups used in the problem. In the cell calculation the function of the fast groups is to provide a source of thermal neutrons and thus only three fast groups have been used. Table III shows how the cell effect is influenced by the number of thermal groups. The use of 2 groups, as long as the group boundary is at ~ 0.14 eV, gives values only somewhat more than 10% lower than the coefficients calculated with 4 or 6 groups. The group structures used are given in Table IV. Recent temperature coefficient calculations have used the 3 fast and 4 thermal groups of Table IV for the cell calculations and the 5 fast—4 thermal group set for diffusion calculations on the entire reactor. The difference in k for a given TRIGA configuration calculated with either 17, 5 or 3 fast groups is less than 0.2%. The 5 fast groups were used mainly for their correlation with the cutoff energies of several foils used for fast flux measurements.

In the past, calculations of reactivity as a function of fuel temperature have been done on the entire reactor with both one- and two-dimensional diffusion codes. Essentially identical temperature coefficient results were obtained from 1D GAZE calculations using 17 fast and 7 thermal groups and 2D GAMBLE calculations with 3 fast and 3 thermal groups. A constant axial buckling value was used for all groups and regions in the one-dimensional cylinder problems.

5.2.3. Uniform Temperature Assumption

TRIGA temperature coefficients have been determined numerically by calculating the change in reactivity associated with a uniform heating of the fuel-moderator elements, with the core water and reflector materials assumed to remain at 23°C. The effects of variation from a uniform temperature for both the cell and the entire core have been investigated with the result that the cell effect is reduced $\sim 5\%$, but this decrease is offset by an increase in the core leakage contribution to the prompt negative temperature coefficient of $\sim 10\%$ when zones of different temperature are incorporated in the cell and reactor calculations.

5.2.4. Doppler Effect

In addition to the effects associated with the neutron spectrum, there are two other smaller contributing effects to the prompt negative temperature coefficient. The first of these is the Doppler broadening of the U^{238} absorption resonances. For the high hydride element this is calculated to contribute about $-2.3 \times 10^{-5}/^{\circ}\text{C}$ at room temperature.

Table III

CELL COMPONENT OF TRIGA PROMPT NEGATIVE
TEMPERATURE COEFFICIENT VERSUS NUMBER
OF THERMAL NEUTRON GROUPS

8.5 wt-% U-ZrH_{1.65} fuel element,
 $\Delta T = 677^{\circ}\text{C}$,

Number of fast groups = 3.

Number of Thermal Groups	$\frac{1}{\eta f} \frac{\Delta \eta f}{\Delta T}$
6	5.49×10^{-5}
4	5.47×10^{-5}
2	4.85×10^{-5}

Table IV
GROUP STRUCTURES USED FOR RECENT TRIGA
TEMPERATURE COEFFICIENT CALCULATIONS

Fast Groups		
5	3	
15.0 MeV to 3.0 MeV 3.0 MeV to 1.5 MeV 1.5 MeV to 0.608 MeV 0.608 MeV to 9.12 keV 9.12 keV to 1.125 eV	15.0 MeV to 0.608 MeV 0.608 MeV to 9.12 keV 9.12 keV to 1.125 eV	
Thermal Groups		
6 (eV)	4 (eV)	2 (eV)
1.125 to 0.65 0.65 to 0.42 0.42 to 0.26 0.26 to 0.14 0.14 to 0.05 0.05 to 0.002	1.125 to 0.42 0.42 to 0.14 0.14 to 0.05 0.05 to 0.002	1.125 to 0.14 0.14 to 0.002

This contribution decreases somewhat with increasing temperature to $-1.8 \times 10^{-5}/^{\circ}\text{C}$ at 1000°C . The second effect is the thermal expansion of the fuel element and consequent expulsion of core water. From the measured void coefficient of TRIGA and the known thermal expansion coefficient of zirconium hydride, this effect should contribute less than $-1 \times 10^{-5}/^{\circ}\text{C}$ to the prompt temperature coefficient. Since the cladding is not bonded to the fuel, this effect will not contribute at all until elevated fuel temperatures are reached. It would seem a reasonable approximation to assume that the sum of the Doppler and fuel element expansion effects contributes a total of $\sim -2 \times 10^{-5}/^{\circ}\text{C}$ to the prompt temperature coefficient, independent of the fuel temperature. This value is about 10 to 15% higher for the $\text{ZrH}_{1.0}$ element.

5.2.5. Total Temperature Coefficient

The calculated temperature coefficients as a function of temperature are shown in Figure 8 for the following two TRIGA reactor configurations:

1. Water reflected core of stainless steel clad 8.5 wt-% U (20% enriched)– $\text{ZrH}_{1.65}$ fuel elements. Stainless steel is 0.020-in.-thick and clad element is 1.475-in.-OD. Core water volume fraction is 1/3. Four water-filled control rod positions in the core are assumed.
2. Core has aluminum clad 8.0 wt-% U (20% enriched)– $\text{ZrH}_{1.0}$ fuel elements 1.475-in.-OD. Clad is 0.030-in.-thick and the core water volume fraction is 1/3. A mixture of 1/3 water and 2/3 aluminum clad dummy elements filled with graphite surrounds the core. This mixture is about 1.5-in.-thick and is surrounded by about one foot of graphite. An infinite water reflector surrounds the graphite. Four water-filled control rod positions in the core are assumed.

Table V shows a breakdown of the relative magnitudes of the major components contributing to the prompt negative temperature coefficient. If fueled control rod followers are used in the core, the contribution from the core inhomogeneities would vanish. This would be true also for void or aluminum followers. When a water reflector is replaced around either core with a graphite reflector, the leakage coefficient becomes about 15% more negative. To note the change in coefficient introduced by leaving boron control rods (or poison followers) in the core, calculations have been done which indicate that the prompt negative temperature coefficient becomes more negative by $\sim 0.1 \times 10^{-5}/^{\circ}\text{C}$ for each percent of rod reactivity worth.

Table V

RELATIVE MAGNITUDE OF CONTRIBUTING COMPONENTS
TO THE PROMPT NEGATIVE TEMPERATURE COEFFICIENT
OF TRIGA REACTORS

Contributing component	Fuel and Cladding	
	ZrH _{1.0} -Al (%)	ZrH _{1.65} -S. S. (%)
Cell effect (change in disadvantage factor)	40	55
Inhomogeneities (control rod locations)	5	5
Doppler broadening of U ²³⁸ resonances	30	20
Leakage from core ^a	25	20

^aThe low-hydride core is assumed to be reflected by graphite radially (TRIGA Mark I and Mark II) whereas the high-hydride core is water-reflected radially (TRIGA Mark F and Mark III). When a water reflector is replaced around either core with a graphite reflector, the leakage coefficient becomes about 15% more negative.

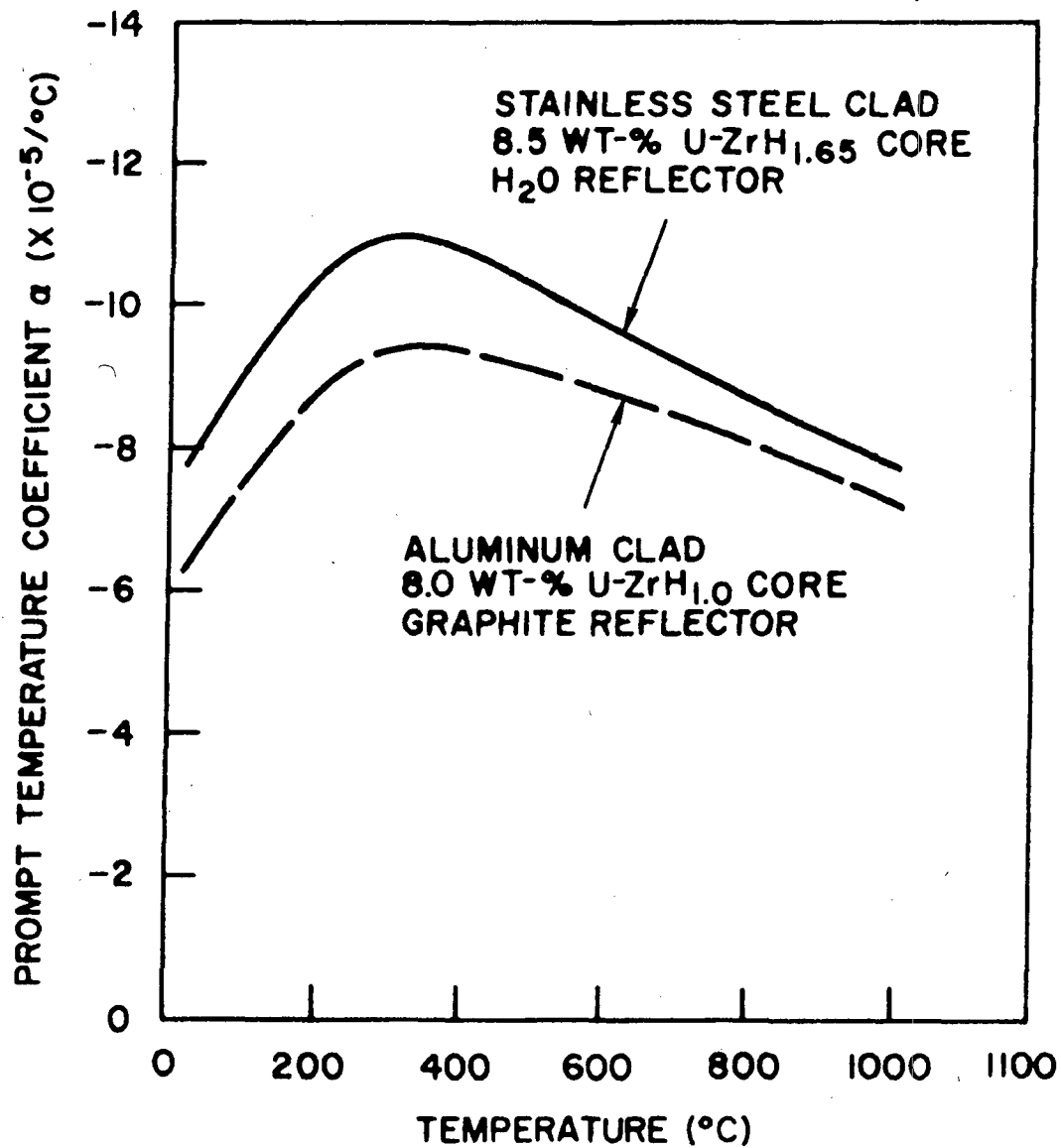


Fig. 8—TRIGA prompt negative temperature coefficient versus average fuel temperature

6. β_{eff} AND ℓ

In order to perform kinetics calculations and interpret experiments, calculations have been made to determine both the effective delayed neutron fraction, β_{eff} , and the prompt neutron lifetime, ℓ . Calculated values are shown in Table VI, where it is seen that the effective delayed neutron fraction was found to be insensitive to the reflector material. Recent die-away measurements in the ATPR confirm the calculated neutron lifetime of 39 μ sec.

The β_{eff} values were derived from reactor calculations where the reactivity was first computed with the prompt fission spectrum alone and then recalculated with the fission spectrum accounting for both prompt and delayed neutrons. The two k_{eff} values thus obtained, k_p and k_t were used in the relationship

$$\beta_{eff} = \frac{k_t(1 + \beta_0)}{k_p} - 1 \quad (1)$$

$$\beta_0 = \text{actual delayed neutron fraction (0.0065).}$$

The prompt neutron lifetime was calculated by the $1/v$ absorber method where a very small amount of boron is homogeneously distributed throughout the system and the resulting change in reactivity is related to the neutron lifetime as follows:

$$\ell = \frac{\delta k_{eff}}{\omega} \quad (2)$$

where ω is related to the boron atomic density by

$$N_{\text{boron}} = \frac{\omega}{\sigma_0 v_0} = 6.024 \times 10^{-9} (\times 10^{24}) \times \omega$$

$$N_B = \text{boron atomic density}$$

$$\omega = \text{integer}$$

$$v_0 = 220,000 \text{ cm/sec}$$

$$\sigma_0 = 755 \text{ barns (2200 meter/sec boron cross section).}$$

The calculation was found to be insensitive to changes in ω between 1 and 100.

Table VI
CALCULATED VALUES OF λ AND β_{eff}

Fuel Element	Reflector	
	Water	Graphite
Stainless steel clad 8 wt-% U-ZrH _{1.7}	$\beta_{\text{eff}} = 0.0070$ $\lambda = 39 \times 10^{-6} \text{ sec}$	$\beta_{\text{eff}} = 0.0070$ $\lambda = 43 \times 10^{-6} \text{ sec}$
Aluminum clad 8 wt-% U-ZrH _{1.0}	$\beta_{\text{eff}} = 0.0073$ $\lambda = 45 \times 10^{-6} \text{ sec}$	$\beta_{\text{eff}} = 0.0073$ $\lambda = 60 \times 10^{-6} \text{ sec}$

7. HEAT CAPACITY

The enthalpy of the TRIGA fuel elements as a function of temperature has been determined from data and fundamental considerations given in a paper by T. B. Douglas,⁽¹⁷⁾ which lends itself fairly readily to different conditions of composition.

Douglas measured the heat content of different samples of zirconium-hydride ranging in hydrogen atomic percent from zero to 55.5. (17-19) From the results of these measurements, and knowing the compositions at the phase boundaries, one can extrapolate the Douglas data to values of x (H atoms per Zr atom) greater than 1.25 (the highest value of the samples studied). This was done to derive an expression for the heat content (above 25°C) of the δ phase ZrH_x , which is approximated well by the following relationship:

$$\begin{aligned} (H - H_{25}) ZrH_x = & 0.03488T^2 + (34.446 + 14.8071(x - 1.65))T \\ & - 882.95 - 370.18(x - 1.65) \text{ joules/mole.} \end{aligned} \quad (3)$$

The temperature is in °C.

The enthalpy of uranium metal was derived from the specific heat data given by Etherington⁽²⁰⁾ and is given by

$$(H - H_{25})U = (0.6483 \times 10^{-4} T^2 + 0.1087 T - 2.758) \text{ joules/gm.} \quad (4)$$

Using the expression

$$\begin{aligned} \rho_{ZrH_x} &= 6.49 - 0.55 H/Zr \text{ gm/cm}^3 \text{ where } H/Zr < 1.6 \text{ and} \\ \rho_{ZrH_{x>1.6}} &= \rho_{ZrH_{1.6}} \end{aligned} \quad (4)$$

to compute a density of 5.610 gm/cm³ for the $ZrH_{1.7}$ and using 18.9 gm/cm³ for the uranium metal density, the volumetric heat content of 8 wt-% U- $ZrH_{1.7}$ alloy was calculated to be

$$\begin{aligned} H - H_{25}(8U-ZrH_{1.7}) &= 2.08 \times 10^{-3} T^2 + 2.12 T - 54.3 \text{ watt-sec/cm}^3 \\ \rho(8U-ZrH_{1.7}) &= 5.944 \text{ gm/cm}^3 \text{ and} \end{aligned} \quad (6)$$

the volumetric specific heat is:

$$C_p = 2.12 + 4.16 \times 10^{-3} T \text{ watt-sec/cm}^3 \text{ } ^\circ\text{C (from } 0^\circ\text{C).} \quad (7)$$

For the 385 cm³ (15-in.-high) fueled portion of a TRIGA fuel element

$$C_p = 857 + 1.60(T - 25^\circ\text{C}) \text{ watt-sec/}^\circ\text{C per element (from } 25^\circ\text{C).} \quad (8)$$

8. FUCHS-NORDHEIM KINETICS MODEL

The kinetic behavior of TRIGA type reactors is primarily determined by their large prompt negative temperature coefficient of reactivity. This coefficient, coupled with the sizeable heat capacity of the fuel-moderator elements allows safe operation with large step insertions of reactivity. The significance of the sizeable heat capacity to store the prompt burst energy is particularly important. There are reactor systems which have large negative temperature coefficients but their usefulness for pulsing operation is limited because of a smaller heat capacity. Under these conditions, temperature limitations on the fuel element, melting or rapid boiling of a homogenized core impose limitations on the energy input of the pulse. Also, if the fuel is lumped in some fashion, such as in a seed-like dispersion through a matrix of moderating material, the effective heat capacity that could be utilized for a pulse might be very small because of the time necessary for heat to transfer from the fuel seeds to the matrix.

The following discussion is given to show qualitatively how the pulsing performance of a reactor is influenced by the parameters which have already been mentioned. The equations given result from the Fuchs-Nordheim space independent kinetics model⁽²¹⁾ in which heat transfer and delayed neutrons are neglected. The applicability of this relatively simple model has been aptly demonstrated both by its performance predictions for the pulses generated to date in the present TRIGA reactor systems, and by studies on the Fuchs-Nordheim model and its application to TRIGA, reported by Scalettar and West.⁽²²⁾ In this model, the reactor parameters which appear are:

ℓ = the prompt neutron lifetime ,

α = the prompt negative temperature coefficient ,

C = the total heat capacity of the core available to the prompt burst energy release ,

$\Delta \bar{T}$ = the change in average core temperature produced by the prompt burst , and

Δk_p = that portion of the step reactivity insertion which is above prompt critical.

If a step insertion of reactivity is made, the reactor power will initially rise exponentially as $\exp(t/\tau)$ where t is time and τ is the reactor period

$$\tau = \frac{\ell}{\Delta k_p} \quad (9)$$

The simplified assumption of constant heat capacity and temperature coefficient (where the temperature coefficient is prompt) leads to a solution where the reactor temperature will rise by $\Delta k_p/\alpha$ until the reactivity insertion beyond prompt critical is just compensated by the fuel temperature rise. The reactor power peaks at this time and then falls to very low values while the core temperature continues to increase to a maximum value double the temperature at peak power

$$\Delta \bar{T} = \frac{2\Delta k_p}{\alpha} = \frac{E}{C} \quad (10)$$

(The peak core temperature is, of course, dependent upon the peak-to-average power ratio in the core.) The total energy release in the burst will be

$$E = \frac{2C \Delta k_p}{\alpha} \quad (11)$$

and the peak power will be

$$P_{\max} = \frac{C(\Delta k_p)^2}{2\ell\alpha} + P_0 \quad (12)$$

where P_0 , the initial power, is usually insignificant.

The following comments are drawn from relationships (Equations 9 through 12):

1. The energies released from pulses vary as the first power of the prompt reactivity insertion.
2. The peak powers for pulses vary as the second power of the prompt reactivity insertion.
3. $P_{\max}$ and E can be increased by increasing C . However, it should be noted that the flux per watt in a core varies inversely with the core volume; thus, increasing the heat capacity by increasing the core size (while offsetting the increase in reactivity with additional neutron absorber) really does nothing to increase the neutron flux resulting from a pulse. High pulse performance is achieved through the use of high heat capacity materials.

4. For a given $\Delta \bar{T}$, the larger the prompt negative temperature coefficient, the larger the prompt reactivity that can be safely compensated. That is to say, large reactivity insertions are necessary to produce limiting core temperatures in a reactor with a large negative temperature coefficient; small reactivities will give the same limiting temperatures in a core with a small α . From both a safety and a convenience point of view the optimum value for α would be such that limiting core temperatures were not reached owing to the use of the Δk in a single control rod or the reactivity changes caused by experiments being maneuvered within or next to the core.

As has been discussed earlier, the heat capacity of the TRIGA fuel is not constant with temperature but can be adequately represented in the form

$$C = C_0 + C_1 T \quad (13)$$

where for 8 wt-% U-ZrH_{1.7}

$$C_0 = 2.10 \text{ watt-sec/cm}^3 \text{ } ^\circ\text{C (at } 0^\circ\text{C) ,}$$

$$C_1 = 4.12 \times 10^{-3} \text{ watt-sec/cm}^3 \text{ } ^\circ\text{C}^2.$$

At 400°C , the above values would give a heat capacity ~ 1.8 times the value at 0°C .

The influence on pulse characteristics caused by variations of this size in the heat capacity can be shown by use of a modified Fuchs-Nordheim model, derived by Scalettar,⁽²³⁾ which allows a linear varying heat capacity. In the modified model, peak power and prompt energy release are calculated from the following:

$$P_{\max} = P_0 + \frac{(\Delta k_p)^2 C_0}{2\alpha t} + \frac{1}{6} \frac{(\Delta k_p)^3 C_1}{\alpha^2 t} \quad (14)$$

$$E = C_0 (T_\infty - T_0) + \frac{C_1}{2} (T_\infty - T_0)^2 \quad (15)$$

T_0 = beginning temperature (at which C_0 is determined) and

$T_\infty - T_0$ = temperature rise during prompt burst.

Because of the variable heat capacity the prompt burst temperature rise is no longer determined by Eq. (10) where $\Delta \bar{T} = 2\Delta k_p/\alpha$ but

is calculated according to the very good approximation

$$\frac{T_{\infty} - T_0}{\Delta T} = -\frac{3}{8}(\sigma - 1) \pm \frac{3}{8} \left((\sigma - 1)^2 + \frac{16}{3}\sigma \right)^{1/2}, \quad (16)$$

where the + sign is used for $\sigma > 1$, and

$$\sigma = \frac{\alpha C_0}{\Delta k_p C_1} \quad (17)$$

It is interesting to compare the pulse characteristics of a typical TRIGA using both the regular and the modified Fuchs-Nordheim model. The following approximate core parameters are used for this purpose:

$$l = 4 \times 10^{-5} \text{ sec}, C_0 \text{ (at } 25^\circ\text{C)} = 8 \times 10^4 \text{ watt-sec/}^\circ\text{C per core},$$

$$\alpha = 1 \times 10^{-4} / ^\circ\text{C}, C_1 = 1.5 \times 10^2 \text{ watt-sec/(}^\circ\text{C)}^2 \text{ per core},$$

$$\Delta k_p = 0.014 \text{ ($3 pulse) and } 0.028 \text{ ($5 pulse)},$$

and both C_0 and $\bar{C}$ will be used in the standard model where,

$$\bar{C} = C_0 + C_1 \frac{\Delta k_p}{\alpha} = C_0 \left(1 + \frac{1}{\sigma} \right) \quad (18)$$

Results of this comparison are shown in Table VII from which it is noted that for the \$3 pulse the constant heat capacity (standard) model gives values of $P_{\max}$ and E about 10% to 15% different from the variable heat capacity model results, and this difference extends to 15 to 30% for a \$5 pulse. The standard model also overestimates core temperatures by as much as 10%. It is also noted that in the modified model the energy release does not vary exactly as the first power of Δk_p , but varies at a slightly steeper rate over our range of interest. $P_{\max}$ also varies at a slightly greater rate than as the second power of Δk_p . Within the range of $\Delta k_p = \sim 4$ to 5 dollars, however, this variation would be difficult to detect experimentally due to the variable core size and/or control rod configurations (and resulting temperature coefficient changes) which are experienced for pulses of sizeably different magnitudes.

In work relating to the effects of a variable temperature coefficient on the Fuchs-Nordheim model, reported by Scalletar and West, (22) it is shown that the variable temperature coefficient characteristic of a TRIGA reactor does not significantly alter the pulse characteristics as calculated by the model with a constant temperature coefficient.

Table VII

COMPARISON OF TRIGA PULSE CHARACTERISTICS USING THE
FUCHS-NORDHEIM POINT KINETICS MODEL WITH BOTH
A CONSTANT AND A VARIABLE HEAT CAPACITY

	\$3 Pulse			\$5 Pulse		
	Modified model	Standard model		Modified model	Standard model	
		$C = C_0$	$C = \bar{C}$		$C = C_0$	$C = \bar{C}$
Δk_p	0.014	0.014	0.014	0.028	0.028	0.028
$P_{\max}$ (Mw)	2,130	1,960	2,475	9,210	7,840	11,950
E (Mw-sec)	26.1	22.4	28.3	59.7	44.8	68.3
$\Delta \bar{T}$ ($^{\circ}\text{C}$)	280	280	280	560	560	560
$T_{\infty} - T_0$ ($^{\circ}\text{C}$)	262	-----	-----	506	-----	-----
$T_{\infty} - T_0 / \Delta T$	0.937	-----	-----	0.904	-----	-----

$$l = 4 \times 10^{-5} \text{ sec} \quad C_0 \text{ (at } 25^{\circ}\text{C)} = 8 \times 10^4 \text{ watt-sec}/^{\circ}\text{C per core}$$

$$\alpha = 1 \times 10^{-4} / ^{\circ}\text{C} \quad C_1 = 1.5 \times 10^2 \text{ watt-sec}/(^{\circ}\text{C})^2 \text{ per core}$$

The Fuchs-Nordheim model as modified for a variable heat capacity has been used to calculate the pulse characteristics of the ATPR for both \$3 and \$5 reactivity insertions. The values obtained are in very close agreement with experimental results. A comparison of the experimental and calculated values is presented in Table VIII.

9. CONCLUSIONS

This report has discussed methods for calculating the prompt negative temperature coefficient, heat capacity, β_{eff} , and l , for TRIGA reactors. These kinetics parameters were used in a modified Fuchs-Nordheim point kinetics model to calculate the pulse characteristics of the ATPR. The calculated characteristics were compared with experimental data and the agreement obtained indicates that the kinetic behavior of U-ZrH systems can be predicted with good accuracy.

Table VIII
TRANSIENT CHARACTERISTICS OF THE ATPR
(100 Fuel Elements)

Prompt temperature coefficient, $\alpha = -11 \times 10^{-5}/^{\circ}\text{C}$. (Changes with step insertion due to varying rod worth remaining in core; value also includes effects of fuel burnup on ATPR core.)

Prompt neutron lifetime, $\ell = 39 \mu\text{sec}$.

Effective delayed neutron fraction, $\beta_{\text{eff}} = 0.007$.

Heat capacity, per element = $857 + 1.60 (T - 25) \text{ watt-sec}/^{\circ}\text{C}$.

All \$5.00 pulses have been performed with water in the A-ring position.

Step Insertion (\$) $\delta k/\beta$	Peak power (Mw)		Energy release (Mw-sec)		Fuel Temperature ^a (°C)				
					At thermocouple			Peak calculated	
						Calculated			
	Experimental	Calculated	Experimental	Aluminum follower in A-ring		H ₂ O follower in A-ring	Aluminum follower in A-ring	H ₂ O follower in A-ring	
5.00	8,400	9,000	54	57	750	775	790±20h	960	1,050
4.50	6,550		45		675				
4.00	4,800	4,900	39	40	585	600	615±15	745	830
3.00	2,000	2,100	24	25	405	420	430±10	520	590

^aAll values are for the prompt burst with no "tail" energy included. The thermocouple is placed 0.3-in. from center of fuel element at the midplane of the core. Calculated temperatures at the thermocouple location have been increased 5% from their adiabatic values to allow for heat flow because the thermocouple did not have a fast enough response time to follow the temperature during the pulse. Peak temperatures occur at the edge of the B-ring elements and are calculated through use of the power distribution within the fuel material.

^bTemperature varies with rotation of the B-ring thermocouple element due to the fact that the thermocouple is located 0.3-in. radially out from the center of the fuel element. There is no significant variation with rotation when an aluminum follower is in the A-ring.

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